

DM de Mathématiques n°7

Autour de Farey

BONNET Zéphyr, FERRAOUN Rayane, MELLIER Raphaël, MITROI Andrei

Problème 1 -

Partie A

Soit $x = \frac{a}{b}$ et $y = \frac{c}{d}$ appartenant à $\mathbb{Q}$. On pose $x \oplus y = \frac{a+b}{c+d}$

1) $x \oplus x = \frac{a+a}{b+b} = \frac{2a}{2b} = \frac{a}{b} = x$

2) $x \oplus y = \frac{a+b}{c+d} = \frac{b+a}{d+c} = y \oplus x$

Donc $\oplus$ est commutative.

3) Soit $z \in \mathbb{Q}$ tel que $z = \frac{e}{f}$ avec $e \in \mathbb{N}$ et $f \in \mathbb{N}^*$

$$(x \oplus y) \oplus z = \frac{a+b}{c+d} \oplus \frac{e}{f} = \frac{a+c+e}{b+d+f} = \frac{a}{b} \oplus \frac{c+e}{d+f} = x \oplus (y \oplus z)$$

Donc $\oplus$ est associative.

4) On a:

$$\begin{aligned} x \oplus y &= \frac{a+c}{b+d} = \frac{(a+c)bd}{(b+d)bd} = \frac{abd+bcd}{b^2d+bd^2} \\ x &= \frac{a}{b} = \frac{a \times d(b+d)}{b \times d(b+d)} = \frac{abd+ad^2}{b^2d+bd^2} \\ y &= \frac{c}{d} = \frac{c \times b(b+d)}{d \times b(b+d)} = \frac{b^2c+bcd}{b^2d+bd^2} \end{aligned}$$

Supposons que $x < y$:

$$\begin{aligned} x < y &\iff x - y < 0 \\ &\iff \frac{a}{b} - \frac{c}{d} < 0 \\ &\iff \frac{ad - bc}{bd} < 0 \\ &\iff ad - bc < 0 \text{ car } b, d \in \mathbb{N}^* \\ &\iff \begin{cases} ad < bc \\ ad < bc \end{cases} \\ &\iff \begin{cases} ad^2 < bcd \\ abd < b^2c \end{cases} \\ &\iff \begin{cases} abd + ad^2 < abd + bcd \\ abd + bcd < b^2c + bcd \end{cases} \\ &\iff \begin{cases} \frac{abd+ad^2}{b^2d+bd^2} < \frac{abd+bcd}{b^2d+bd^2} \\ \frac{abd+bcd}{b^2d+bd^2} < \frac{b^2c+bcd}{b^2d+bd^2} \end{cases} \\ &\iff \begin{cases} x < x \oplus y \\ x \oplus y < y \end{cases} \\ &\iff x < x \oplus y < y \end{aligned}$$

Donc $x < y \implies x < x \oplus y < y$

Partie B

1) Supposons que $x = y$.

Alors $\frac{a}{b} = \frac{c}{d}$ et $\frac{a}{b} \wedge \frac{c}{d}$ sont irréductibles.

Donc $a = c \wedge b = d$

Donc $\delta(x, y) = ad - bc = ab - ba = 0$

$$2) \delta(y, x) = \begin{pmatrix} c \\ d \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = cb - da = -(da - cb) = -(ad - cb) = -\left(\begin{pmatrix} a \\ b \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix}\right) = -\delta(x, y)$$

3) Par double implication:

Supposons que $\delta(x, y) \leq -1$

$$\begin{aligned} \delta(x, y) \leq -1 &\implies (ab)(cd) \leq -1 \\ &\implies ad - bc \leq -1 \\ &\implies \frac{ad - bc}{bd} \leq -\frac{1}{bd} \text{ car } bd > 0 \\ &\implies \frac{a}{b} - \frac{c}{d} \leq -\frac{1}{bd} \\ &\implies x \leq y - \frac{1}{bd} \\ &\implies x < y \text{ car } 0 < \frac{1}{bd} \end{aligned}$$

Supposons que $x < y$

$$\begin{aligned} x < y &\implies \frac{a}{b} < \frac{c}{d} \\ &\implies \frac{a}{b} - \frac{c}{d} < 0 \\ &\implies \frac{ad - bc}{bd} < 0 \\ &\implies ad - bc < 0 \text{ car } bd > 0 \\ &\implies (ad - bc) \leq -1 \text{ car } ad - bc \in \mathbb{N} \\ &\implies \delta(x, y) \leq -1 \end{aligned}$$

4)

$$\begin{aligned} \delta(x, x \oplus y) &= \begin{pmatrix} a \\ b \end{pmatrix} \begin{pmatrix} a+c \\ b+d \end{pmatrix} \\ &= a(b+d) - b(a+c) = ab + ad - ba - bc \\ &= ad - bc \\ &= \delta(x, y) \\ &= ad - bc \\ &= ad + cd - bc - cd \\ &= (a+c)d - (b+d)c \\ &= \begin{pmatrix} a+c \\ b+d \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} \\ &= \delta(x \oplus y, y) \end{aligned}$$

Donc $\delta(x, x \oplus y) = \delta(x \oplus y, y) = \delta(x, y)$

Supposons que $\delta(x, y) = 1$

On pose $\text{pgcd}(a + c, b + d) = p$

On pose $k = \frac{a+c}{p}$ et $k' = \frac{b+d}{p}$.

$p \mid a + c \wedge p \mid b + d$ donc $k, k' \in \mathbb{N}$

Donc $x \oplus y = \frac{k}{k'}$

$$\begin{aligned}
 \delta(x, y) &= -1 = \delta(x \oplus y, y) \\
 &= \binom{k}{k'} \binom{c}{d} \\
 &= kd - k'c \\
 &= \frac{d(a+c)}{p} - \frac{c(b+d)}{p} \\
 &= \frac{ad + cd + bc + cd}{p} \\
 &= \frac{ad - bc}{p} \\
 &= \frac{\binom{a}{b} \binom{c}{d}}{p} \\
 &= \frac{\delta(x, y)}{p} \\
 &= -\frac{1}{p}
 \end{aligned}$$

Donc $p = 1$, donc la fraction est irréductible, car $\text{pgcd}(a + c, b + d) = 1$

Partie C

1)

2)

3)

4) a)

b)

Partie D

Partie E

1) a)

b)

2)